lineage; however, like the Arduino, Processing gets its power and usability from carefully-written libraries of code. Processing has gained wide acclaim and has been used in commercials for Nike and Hewlett-Packard, music videos for Modest Mouse and Radiohead, and all manner of art installations everywhere. It is a really good language for visualizing data, creating immersive and interactive graphics, and generating moving images.

To get started using Processing with the Arduino, we need to download Processing and the Arduino Library for Processing from `http://processing.org/download` and `www.arduino.cc/playground/Interfacing/Processing` respectively. Figure 11-13 shows the arduino_output sketch included with this last download.

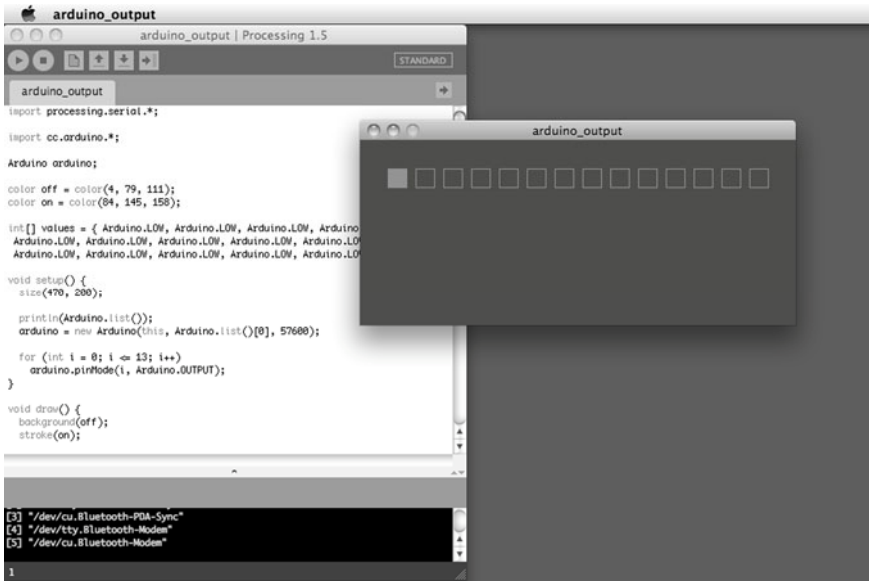



***Figure 11-13.*** *Simple Processing sketch*

If Processing looks familiar, that's because the Processing development environment is the same one that the Arduino environment is built from. Things are a little different and the language will take some getting used to, but once you have a firm grasp of programming the Arduino, it's not overly difficult to get up to speed on Processing.

That is only the briefest introduction to working with Processing. For more information on using the Firmata, Arduino, Processing triumvirate check out the Arduino Playground site, as well as the (somewhat outdated) tutorial from Golan Levin at `www.flong.com/blog/2010/installing-arduino-with-firmata-for-maxmsp-and-processing-in-osx`. The creators of Processing, Casey Reas and Ben Fry, have also done outstanding jobs writing about the language with their books *Getting Started with Processing* (O'Reilly Media, 2010) and *Processing: A Programming Handbook for Visual Designers and Artists* (MIT Press, 2007). Both are great books to pick up if you are interested in learning more about Processing.